

Ex. No: 03

Histogram Equalization Using OpenCV (Grayscale & Color Images)

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Write a Python program using OpenCV to perform histogram equalization on the given grayscale image "parrot.jpg".

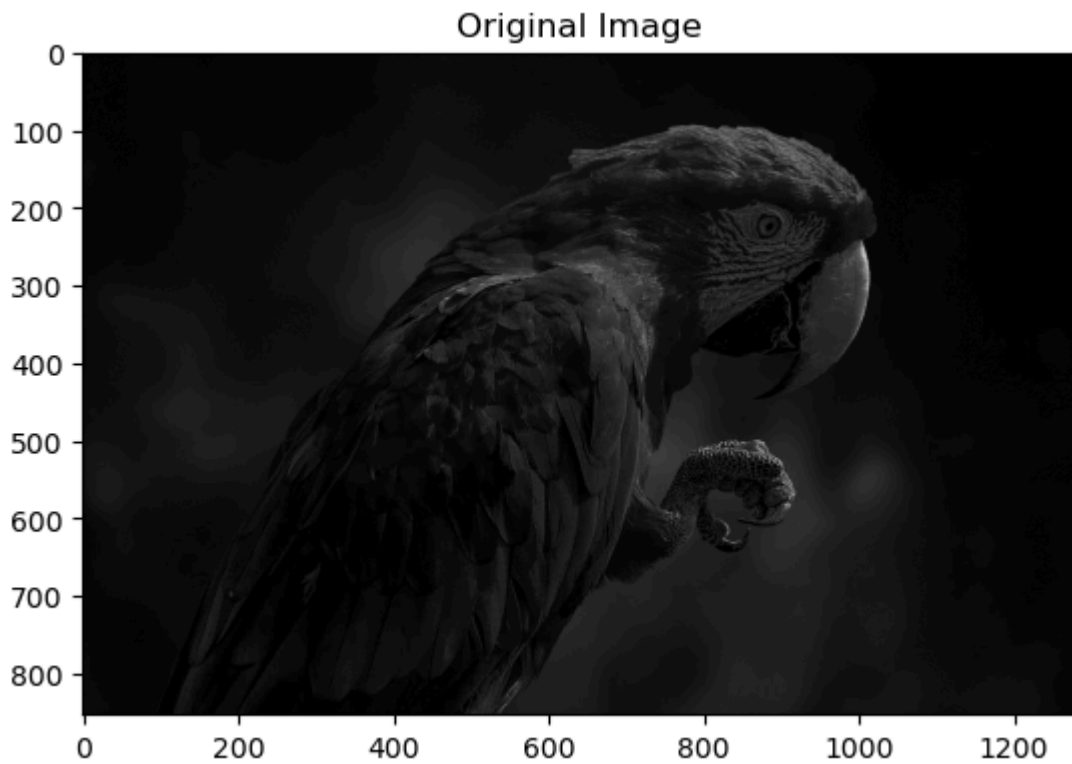
The program should:

- Import the required libraries (**OpenCV, NumPy, Matplotlib**).
- Read the image "**parrot.jpg**" in **grayscale** format.
- Display the grayscale image and plot its histogram.
- Perform **histogram equalization** on the grayscale image using `cv2.equalizeHist()` to enhance its contrast.
- Using Matplotlib, display the following in a **2 × 2 grid layout**:
 - Original Grayscale Image
 - Histogram of Original Image
 - Enhanced (Equalized) Image
 - Histogram of Enhanced Image

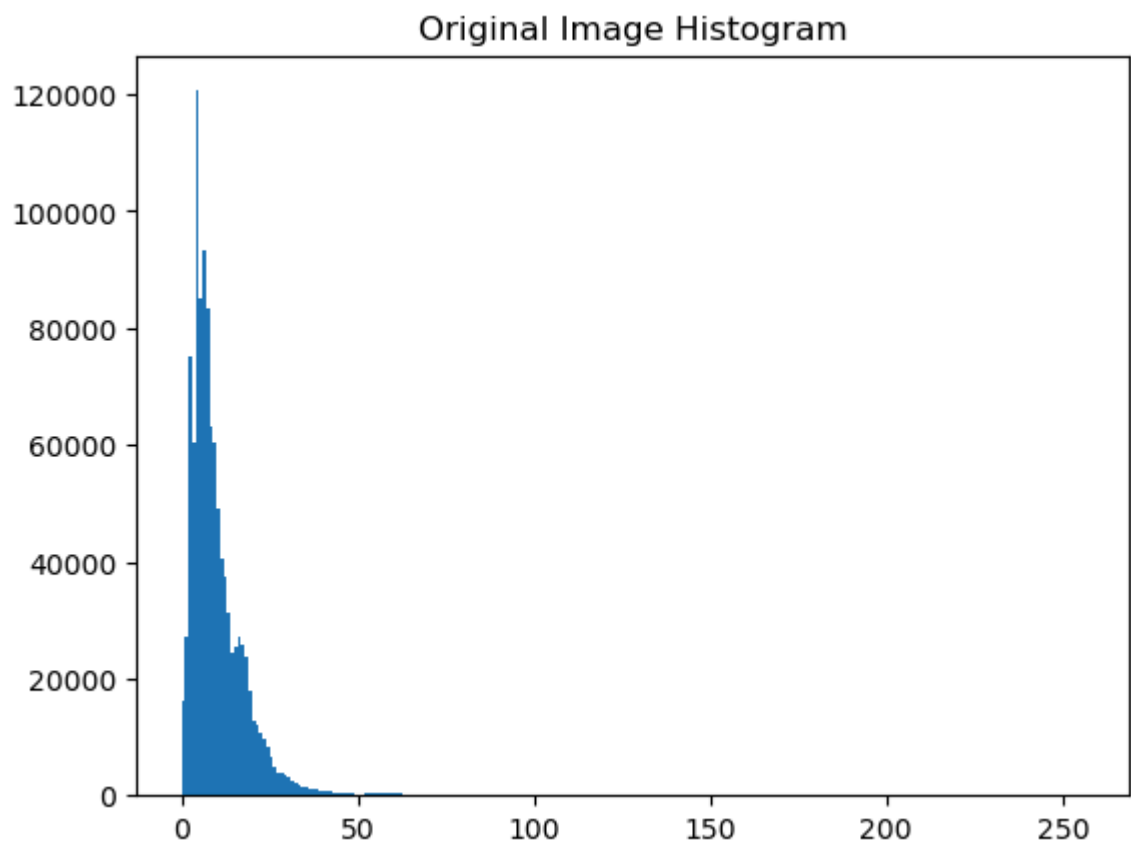
```
In [14]: # Import required libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread('parrot.jpg', cv2.IMREAD_GRAYSCALE)

plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.show()
```

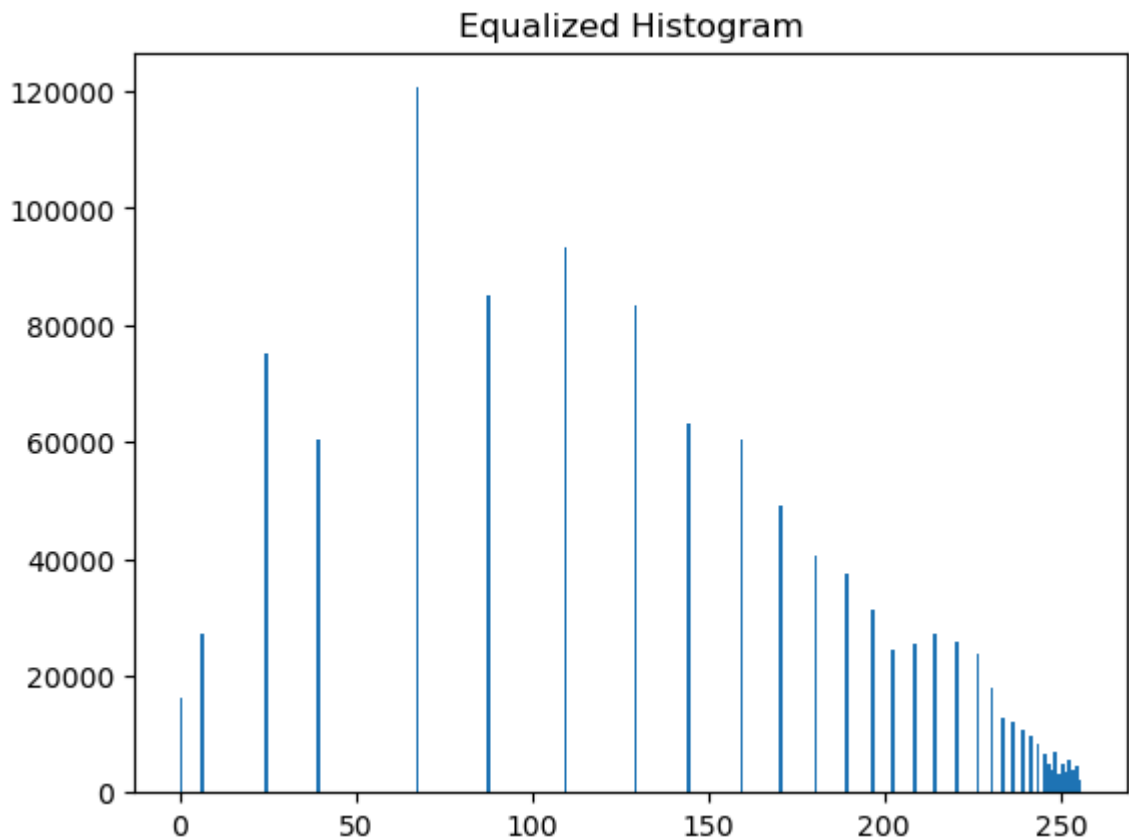


```
In [15]: plt.hist(img.ravel(), 256, range=[0,256])  
plt.title('Original Image Histogram')  
plt.show()
```

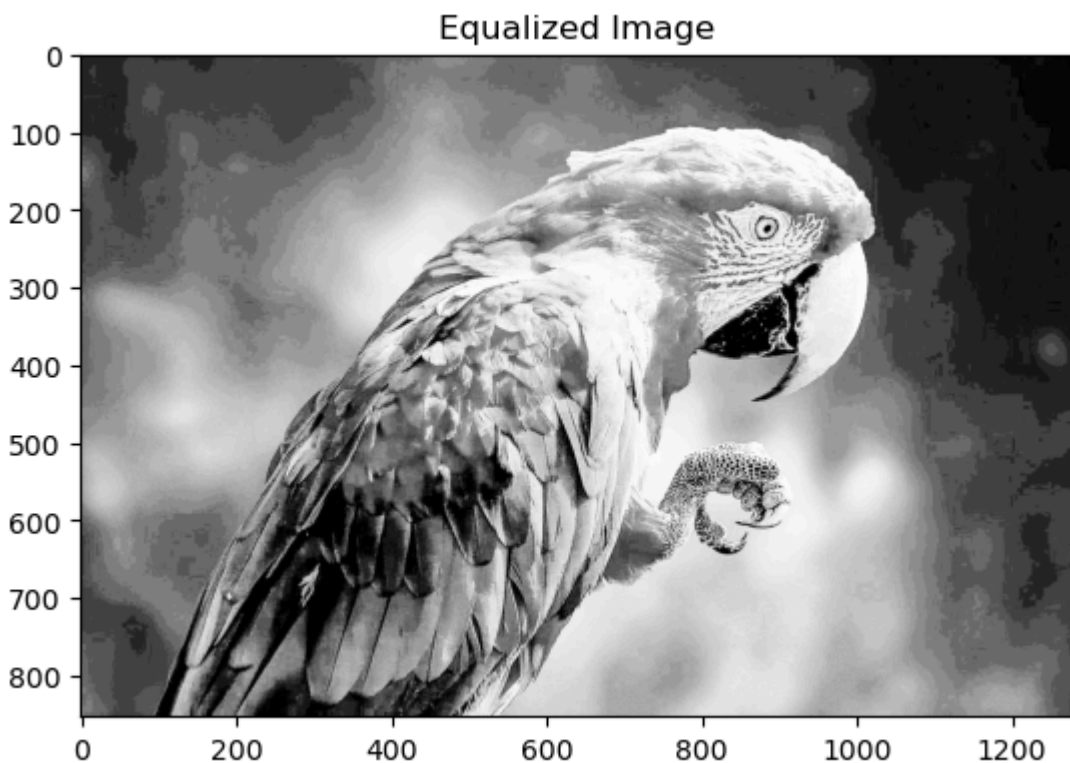


```
In [16]: img_eq = cv2.equalizeHist(img)
```

```
In [17]: plt.hist(img_eq.ravel(), 256, range=[0,256])  
plt.title('Equalized Histogram')  
plt.show()
```



```
In [19]: plt.imshow(img_eq, cmap='gray')  
plt.title('Equalized Image')  
plt.show()
```



```
In [13]: # Display [1] the Original Image (Gray Image) and its Histogram, and [2] the Enh
```

Write a Python program using OpenCV to perform color image enhancement through histogram equalization in the

HSV color space on the given image "parrot.jpg".

The program should:

- Import the required libraries (**OpenCV, NumPy, Matplotlib**).
- Read the color image "**parrot.jpg**".
- Plot the **histogram of the original color image** (use the **B, G, and R** channels).
- Convert the image from **BGR to HSV** color space.
- Apply **histogram equalization** to one specific channel to improve brightness and contrast using `cv2.equalizeHist()` .
- Convert the enhanced HSV image back to **BGR** format.
- Using Matplotlib, display the following in a **2 × 2 grid layout**:
 - Original Color Image
 - Histogram of Original Image (BGR channels)
 - Enhanced Color Image
 - Histogram of Enhanced Image (BGR channels)

```
In [29]: # Import required libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
In [30]: # Read the color given parrot.jpg image.

img = cv2.imread('parrot.jpg', cv2.IMREAD_COLOR)
```

```
In [31]: # Plot the histogram of colour image
img_hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
```

```
In [32]: # Convert to HSV.

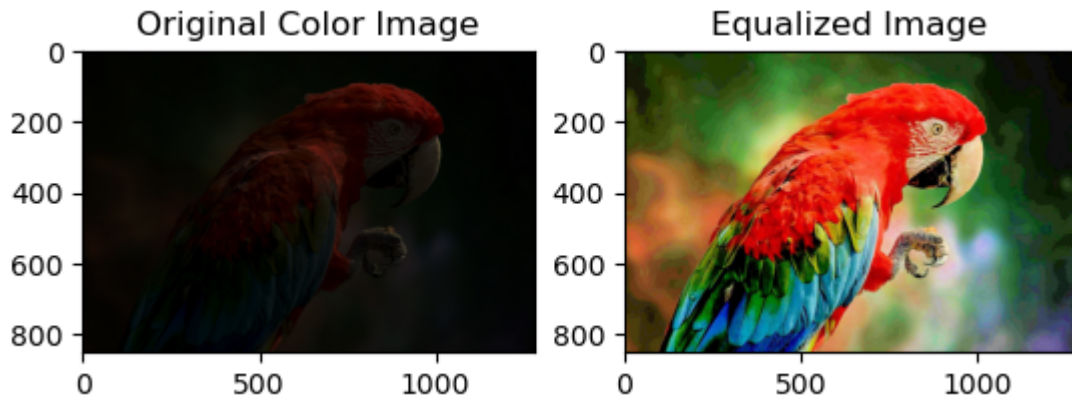
img_hsv[:, :, 2] = cv2.equalizeHist(img_hsv[:, :, 2])
```

```
In [33]: # Perform histogram equalization
img_eq = cv2.cvtColor(img_hsv, cv2.COLOR_HSV2BGR)
```

```
In [34]: # Convert back to BGR format
plt.subplot(121)
plt.imshow(img[:, :, :-1])
plt.title('Original Color Image')

plt.subplot(122)
plt.imshow(img_eq[:, :, :-1])
plt.title('Equalized Image')

plt.show()
```



```
In [28]: # Display
plt.figure(figsize=[12,10])

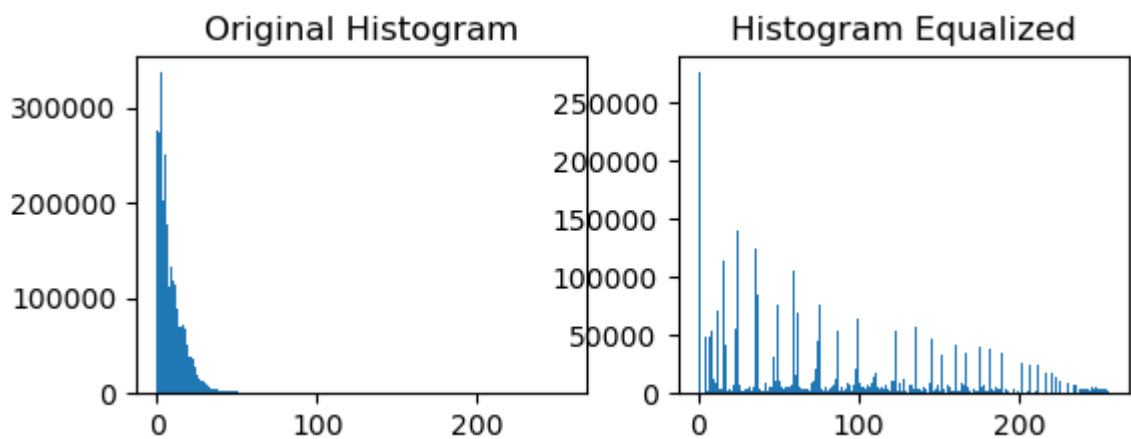
plt.subplot(221)
plt.imshow(img[:, :, :-1])
plt.title('Original Color Image')

plt.subplot(222)
plt.imshow(img_eq[:, :, :-1])
plt.title('Equalized Image')

plt.subplot(223)
plt.hist(img.ravel(), 256, range=[0,256])
plt.title('Original Histogram')

plt.subplot(224)
plt.hist(img_eq.ravel(), 256, range=[0,256])
plt.title('Histogram Equalized')

plt.show()
```



In []: